Modern Academy
for Engineering and Technology in Maadi
Electronic Engineering and Communications
Technology Department.

Academic Year: 2022-2023
Semester: Spring
Exam Date: 11 / 6 / 2023
Level: 4th Level

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (ELCN425) Digital Signal Processing
Spec.: Communications & Computer
Examiners: Dr. Samir M. Kamal and Dr. Sara M. Hassan
Time: 2 Hours

Number of Pages: One | Number of Questions: 4 | Answer All Questions

Question 1 [Los: 1, 3] **(10 points)**

a- In detail, what are the main characteristics of the discrete time sinusoidal signals? 2

b- For the signal $x(t) = 10\cos(1000\pi t)$:

(i) If it is sampled with Fs = 2000 sample/sec, is it possible to reconstruct this signal back from its samples successfully? 2

(ii) Find the Nyquist frequency, F_N, for this signal. 1

(iii) Using the obtained Nyquist frequency, F_N find an expression for the corresponding sequence, x(n).

c- For the following sequence: x(n) = {1, 0, 1, 0}:

(i) Using the FFT algorithm, find the frequency components of this sequence. 2

(ii) Check your results obtained in part (i) by calculating x(n) using the IDFT method. 2

(iii) What is the computational saving if we use FFT algorithm instead of DFT method. 1

Question 2 [Los: 3, 4] **(10 points)**

a- Define (using equations and/or figures) the following items: Correlation cofficent and Convolution operation.

b - Given two sets of samples $x_1(n) = \{2, 2, 0, 1\}$ and $x_2(n) = \{1, 2, 0, 1\}$.

(i) Find the linear cross- correlation between $x_1(n)$ and $x_2(n)$.

(ii) Plot the obtained linear correlation v.s. the lag number, j.

(iii) Find their convolution, y(n).

(iv) Plot the obtained convolution v.s. n.

Question 3 [Los: 1, 4] **(10 points)**

a- For LTIS define (using equations, figures and examples): Linearity, Causality & ROC.

b- Find the discrete-time sequence, h(n), with the following transfer function:

$$H(z) = \frac{z^2}{(z-0.5)(z-1)^2}$$

c- For the transfer function given in part (b):

(i) Find the difference equation of this filter, y (n).

(ii) Represent the filtering operation in a block diagram form.

(iii) Using the Residue method find "4" values for the impulse response of this system.

Question 4 [Los: 2, 4] **(10 points)**

a- Compare between the FIR and IIR filters.

b- What are the different steps of the digital filter design?

c- The transfer function for an FIR filter is given by:

$$H(z) = 1 - 1.3435z^{-1} + 0.9025z^{-2}$$

(1) Draw the realization block diagram for each of the following cases:

(a) Transversal structure;

(b) A two-stage lattice structure.

(2) Calculate the values of the coefficients for the lattice structure.

Good Luck